Homework 5: Chapters 5 & 6

Mathematical Models

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Due

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Chapter 5

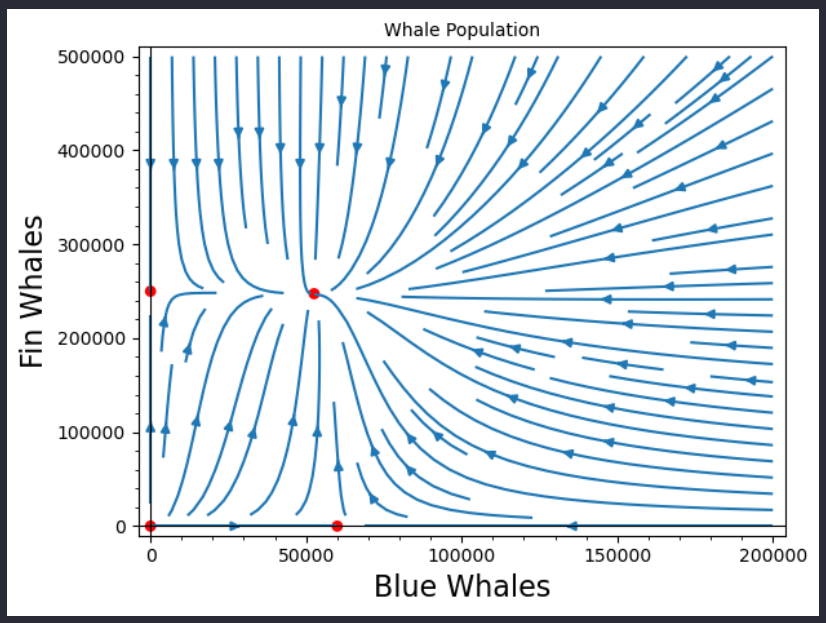

Figure 1: Vector field for Exercise 3, part (a) in Chapter 5 with equilibriums in red.

Blue Whales	Fin Whales	Eigen 1	Eigen 2	Type	Stability
0	0	0.020	0.051	Source	Unstable
60000	0	0.051	−0.020	Saddle	Unstable
0	250000	0.018	−0.051	Saddle	Unstable
52579	247371	−0.051	−0.020	Sink	Stable

Table 1: Each equilibrium's location in Figure 1 with type and stability specified.

3. Reconsider Exercise 6 of Chapter 4. Assume that the catchability coefficient $q = 10^{-5}$ and that the level of effort is $E = 3000$ boat-days per year.

(a) Sketch the vector field for this model. Determine the location of each equilibrium in the state space.

> *Solution.* Figure 1 gives us the vector field for this model. Table 1 gives each location of each equilibrium.

(b) Use the eigenvalue method to test the stability of each equilibrium in the state space.

> *Solution.* Table 1 gives the stability of each equilibrium.

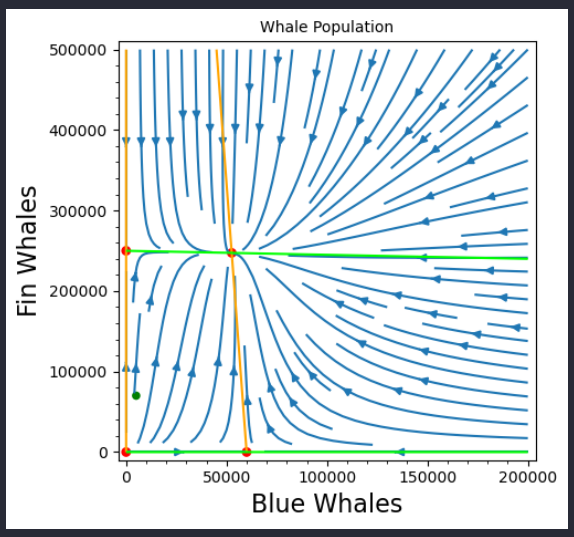

Figure 2: Phase portrait for Exercise 3, part (c) in Chapter 5 with x-nullclines in green, and y-nullclines in orange. The green dot is added to visualize the population for part (d).

(d) Sketch the complete phase portrait for this model, using the results of parts (a) and (c).

> *Solution.* Phase portrait pictured in Figure 2.

(e) Given the current estimates of 5000 blue whales and 70,000 fin whales, what does this model predict about the future of the two species?

> *Solution.* The model predicts that the two species will survive. By following the streamplot in Figure 2, we can see that both species will trend toward the stable equilibrium at $(52579, 247371)$.

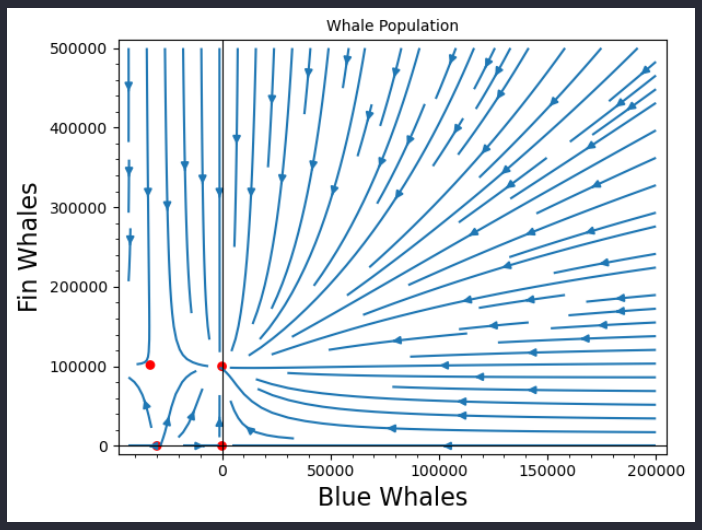

Figure 3: Vector field for Exercise 4, part (a) in Chapter 5 with equilibriums in red.

Blue Whales	Fin Whales	Eigen 1	Eigen 2	Type	Stability
0	0	-0.0098	0.020	Saddle	Unstable
-30000	0	0.020	0.0098	Source	Unstable
0	100000	-0.011	-0.020	Sink	Stable
-33050	101652	-0.020	0.011	Saddle	Unstable

Table 2: Each equilibrium's location in Figure 3 with type and stability specified.

4. Repeat Exercise 3, but now assume a level of effort of $E = 6000$ boat-days per year.

   (a) Sketch the vector field for this model. Determine the location of each equilibrium in the state space.

   Solution. Figure 3 gives us the vector field for this model. Table 2 gives each location of each equilibrium.

   (b) Use the eigenvalue method to test the stability of each equilibrium in the state space.

   Solution. Table 2 gives the stability of each equilibrium.

   (d) Sketch the complete phase portrait for this model, using the results of parts (a) and (c).

   Solution. Phase portrait pictured in Figure 4.

   (e) Given the current estimates of 5000 blue whales and 70,000 fin whales, what does this model predict about the future of the two species?

   Solution. The model predicts that blue whales will quickly go extinct. It also predicts that the fin whales will reach a stable equilibrium at $(0, 100000)$.

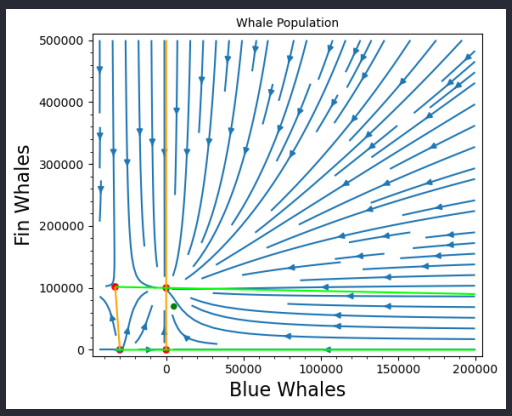

Figure 4: Phase portrait for Exercise 4, part (c) in Chapter 5 with x-nullclines in green, and y-nullclines in orange. The green dot is added to visualize the population for part (d).

Chapter 6

w	**Winner**	**Remaining**	**Time (days)**
0.2	Red	2.07	48.9
0.5	Red	2.8	19.5
0.75	Red	3.03	13.0
0.9	Red	3.11	10.9

Table 3: Simulation results for varying values of w with $\lambda = 3$ in Eq. 6.20.

1. Reconsider the war problem of Example 6.1. In this problem we explore the effects of weather on combat. Bad weather and poor visibility decrease the effectiveness of direct fire weapons for both sides. The effectiveness of indirect fire weapons is relatively unaffected by the weather. We can represent the effects of bad weather in our model as follows. Let w denote the decrease in weapon effectiveness caused by bad weather conditions, and replace the dynamical system in Eq. (6.3) by

$$\Delta x_1 = -w\lambda(0.05)x_2 - \lambda(0.005)x_1x_2$$
$$\Delta x_2 = -w0.05x_1 - 0.005x_1x_2$$

(a) Use a computer implementation of the algorithm in Fig. 6.2 to simulate the discrete–time dynamical system in Eq. (6.20) in the case $\lambda = 3$. Assume that adverse weather conditions cause a 75% decrease in weapon effectiveness for both sides ($w = 0.25$). Who wins the battle, and how long does it take? How many divisions of troops remain on the winning side?

> *Solution.* With the given parameters of $\lambda = 3$ and $w = 0.25$, the simulation returns that Red wins the battle, they have 2.28 divisions remaining, and the battle lasted 39.0 days.

(b) Repeat your analysis for each of the cases $w = 0.1$, 0.2, 0.5, 0.75, and 0.9, and tabulate your results. Answer the same questions as in part (a).

> *Solution.* The tabulated results are presented in Table 3. In the table, we can see that as w increases, so does red's efficiency. In fact, for $w = 0.9$, red almost quintuples their time efficiency, and an additional division survives in comparison to $w = 0.2$.

(c) Which side benefits from fighting in adverse weather conditions? If you were the blue commander, would you expect red to attack on a sunny day or a rainy day?

> *Solution.* It appears that red gains a significant advantage from poor weather. The blue commander should expect an attack on a rainy day.

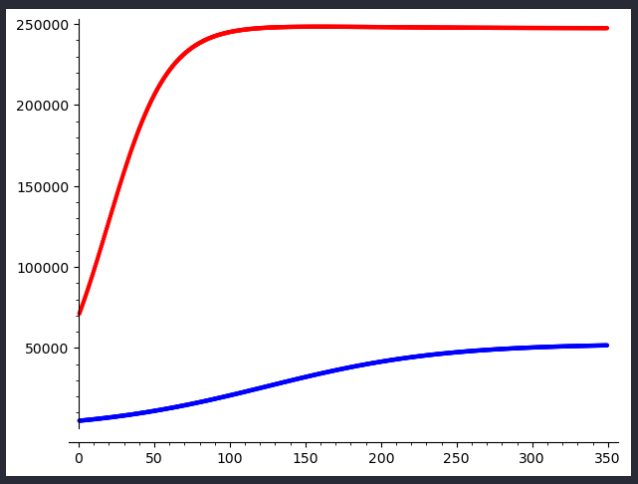

Figure 5: Blue whales (blue line) and fin whales (red line) plotted over time for Exercise 8, part (a) in Chapter 6.

8. Reconsider Exercise 6 in Chapter 4, and assume $\alpha = 10^{-8}$. Assume that the current population levels are $B = 5{,}000$ and $F = 70{,}000$.

(a) Use a computer implementation of the simple algorithm used in Section 6.2 to determine the effect of harvesting. Assume $E = 3{,}000$ boat–days per year. What happens to the two species of whales over the long term, according to this model? Do both species of whales grow back, or will one or both species become extinct? How long does this take?

Solution. Given a harvesting rate of $E = 3000$ boat-days per year and a starting population of 5,000 and 70,000 for blue and fin whales respectively, we can expect both whales will grow back. They will reach a stable equilibrium around the population of $(52579, 247371)$. However, it will take approximately 349.5 years to reach these levels. We also notice in Figure 5 that the fin whales grow back much quicker than the blue whale.

(d) Repeat part (c) for the specific cases $\alpha = 10^{-9}$, 10^{-8}, 10^{-6}, and 10^{-5}, and tabulate your results. Discuss the sensitivity of your conclusions to the extent of interspecies competition.

Solution. From Table 4, we can see that there is a general trend: As we increase E, the upper bound for both whale populations decreases. Then, for $\alpha = 10^{-6}$ and 10^{-5}, we also see that the fin whales consistently break out the blue whales, but the competition does not affect the upper limit for the whale populations. In general, we see that an increase in the competition factor leads to the extinction of blue whales. For the entries with times exceeding 1,000 years, further analysis shows that the fin whales live, but the blue whales are going extinct very slowly.

Blue Whales	**Fin Whales**	**Time (years)**	α	E
117970	349415	199	10^{-9}	1000
108665	344618	209	10^{-8}	1000
0	350000	10000	10^{-6}	1000
0	209890	27.0	10^{-5}	1000
88112	299564	245.50	10^{-9}	2000
80126	296032	260	10^{-8}	2000
0	300000	10000	10^{-6}	2000
0	205809	34.5	10^{-5}	2000
58255	249712	326	10^{-9}	3000
51579	247447	349.50	10^{-8}	3000
0	250000	10000	10^{-6}	3000
0	203739	50.5	10^{-5}	3000
28404	199859	502.50	10^{-9}	4000
23039	198860	550.50	10^{-8}	4000
0	200000	10000	10^{-6}	4000
0	200000	10000	10^{-5}	4000
550	149997	3412.5	10^{-9}	5000
0	150000	10000	10^{-8}	5000
0	150000	10000	10^{-6}	5000
0	150000	10000	10^{-5}	5000

Table 4: Blue and fin whales competition with competition coefficient, α, and effect of harvesting, E for Exercise 8, part (d) in Chapter 6.